

Parametric Equations of Lines

General Parametric Equations

In this part of the unit, we are going to look at parametric curves. This simple idea is that a point moving in space traces out a path over time.

The point (x, y, z) depends on a variable t . Since the position depends on t , we write:

$$x = x(t), \quad y = y(t), \quad z = z(t)$$

Parametric Equations of Lines

Later we will look at general curves. Right now, let us suppose our point moves on a line.

The basic data we need in order to specify a line are a point on the line and a vector parallel to the line. That is, we need a point and a direction.